

Stack & Queue Implementation using Structures (Self-Learning)

Q. Why is it advantageous to implement Stacks and Queues using Structures in C? Provide a C program demonstrating the implementation of a Stack and a Queue where their components are encapsulated within a structure.

> **Definition to Remember**

1. Advantages of Using Structures

- **Encapsulation:** Data array and control variables are logically grouped together.

2. Stack Implementation using Structure

```
```c
```

```
// Define the Stack Structure
```

```
// Initialize stack
```

```
// Push operation
```

### 3. Queue Implementation using Structure

```
```c
```

```
// Define the Queue Structure
```

typede

```
// Initialize queue
```

void in

```
// Enqueue operation
```

void en

```
---
```

```
> Must-Write Points (for 10 marks)
```

> 1. St

```
---
```

```
> Quick Recall
```

> `Imp

Pass-k